# *Navier-Stokes equations in gas dynamics: Green's function, singularity, and well-posedness*

Shih-Hsien Yu joint work with Tai-Ping Liu

Institute of Mathematics
Academia Sinica, Taiwan

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• Well-posedness of weak solution is a subtle subject.

• Well-Posedness may not valid for weak solution.

Non-uniqueness results for weak solutions of incompressible Euler equations, using the technique of convex integration:

Scheffer, V. An inviscid flow with compact support in space-time. J. Geom. Anal. 3(4), 343-401 (1993).

De Lellis C, Szkelyhidi L. On admissibility criteria for weak solutions of the Euler equations. Arch. Ration. Mech. Anal. 195 (2010), no. 1, 225-260.

• There well-posedness theories for system of hyperbolic conservation laws.

The existence theory is done by Glimm scheme:

Glimm, James. Solutions in the large for nonlinear hyperbolic systems of equations. Comm. Pure Appl. Math. 18 (1965), 697-715.

The continuous dependence on the initial data is done by the functional approach:

Liu, T.-P. The deterministic version of the Glimm scheme. Comm. Math. Phys. 57 (1977), no. 2, 135-148.

Liu, T.-P.; Yang, T. A new entropy functional for a scalar conservation law. Comm. Pure Appl. Math. 52 (1999), no. 11, 1427-1442.

By the homotopy approach: Bressan, A. Contractive metrics for nonlinear hyperbolic systems. Indiana Univ. Math. J. 37 (1988), no. 2, 409-421.

On the other hand, there is no proof of well-posedness theory for solutions obtained by the technique of compensated compactness. **The well-posedness theory is done within the Glimm solution algorithm.**

• Weak solution for compressible Navier-Stokes equation:
(No proof of well-posedness)

Hoff, D.,Global solutions of the Navier-Stokes equations for multidimensional compressible flow with discontinuous initial data. JDE, 120 (1995) 215-254.

Lions, P.L., Mathematical Topics in Fluid Mechanics: Volume 2: Compressible Models. Oxford Lecture Series in Mathematics and its Applications, 10. (1996).

Feireisl, E. Dynamics of Viscous Compressible Fluids. Oxford Lecture Series in Mathematics and its Applications, 26. (2004).

• The purpose of this talk is to present another example of well-posedness theory for the weak solutions, besides that for system of hyperbolic conservation laws. We consider the simplest system of viscous conservation laws in physics, the Navier-Stokes equations in gas dynamics.

• Compressible NS equation in the Lagrangian coordinate

$$\begin{cases} v_t - u_x = 0, \\ u_t + p(v)_x = \kappa(u_x/v)_x, \\ p(v) = v^{-\gamma}, \gamma \in (1, 5/3), \kappa = 1, \end{cases} \qquad \begin{cases} v: \text{ volume per unit mass,} \\ u: \text{ particle velocity,} \\ p(v): \text{ pessure with } \gamma \text{ law,} \\ \kappa: \text{ Dissipation constant.} \end{cases}$$

• Nash, J., *Le probleme de Cauchy pour les equations differentielles d'un fluide general,* Bull. Soc. Math. France, 90 (1962) 487-497.

• Kanel, Ya . I., *On a model system of equations for one-dimensional gas motion*, Diff. Eq. (in Russian) 4 (1968) 721-734.

• Kazhikhov, A. V. and Shelukhin, V . V ., *Unique global solution in time of initial boundary value problems for one-dimensional equations of a viscous gas*, Prikl. Mat. Mech., 41 (1977) 282-291.

• Itaya , N., *On the Cauchy problem for th e system of fundamental equations describing the movement of compressible viscous fluid*, Kodai M ath. Sem . Rep., 23 (1971) 60-120.

• Matsumura, A., and Nishida, T., *The initial value problem for the equations of motion of viscous and heat conductive gases*. J. Math. Kyoto Univ. 20-1 (1980) 67-104.

• Hoff, D.,*Global well-posedness of the Cauchy problem for the Navier-Stokes equations of nonisentropic flow with discontinuous initial data,* Journal of differential equations, JDE, 95 (1992) 33-74.

• Lions, P.L., *Mathematical Topics in Fluid Mechanics*: Volume 2: Compressible Models. Oxford Lecture Series in Mathematics and its Applications, 10. (1996).

• Feireisl, E. *Dynamics of Viscous Compressible Fluids*. Oxford Lecture Series in Mathematics and its Applications, 26. (2004).

# *Some remarks on weak solutions.*

• NS.
$$\begin{cases} v_t - u_x = 0, \\ u_t + p(v)_x = \kappa(u_x/v)_x. \end{cases}$$

• The weak solution:
$\int_0^\infty \int_{\mathbb{R}} -(\phi_t v - \phi_x u)dxdt = 0,$
$\int_0^\infty \int_{\mathbb{R}} -(\psi_t u + \psi_x(p(v) - \kappa u_x/v))dxdt = 0$ for all test functions $\phi$,$\psi$.

**Remark:** This formulation allows $v(x,t)$ and $u_x(x,t)$ to contain discontinuities; and the discontinuities can survives for $t > 0$ due to the hyperbolic-parabolic structure.

• Artifical viscosity coefficient problem:
$$\begin{cases} v_t - u_x = \kappa v_{xx}, \\ u_t + p(v)_x = \kappa u_{xx}. \end{cases}$$

• The weak solution:

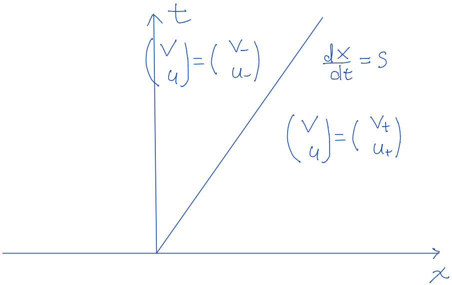
$\int_0^\infty \int_{\mathbb{R}} -(\phi_t v - \phi_x u + \kappa\phi_{xx} v)dxdt = 0,$
$\int_0^\infty \int_{\mathbb{R}} \Big(-(\psi_t u + \psi_x p(v)) - \kappa\psi_{xx} u\Big)dxdt = 0$ for all test functions $\phi$,$\psi$.

**Remark:** $u(x,t)$ and $v(x,t)$ become continuous for $t > 0$.

# *Rankine-Hugoniot condition for discontinuities in NS*

- NS in the conservative form.

$$\begin{pmatrix} v \\ u \end{pmatrix}_t + \begin{pmatrix} -u \\ p(v) - \kappa u_x / v \end{pmatrix}_x = \vec{0}.$$




- Rankine-Hugoniot Condition.

$$\begin{cases} s[v] = -[u], \\ s[u] = [p(v) - \kappa u_x], \\ \text{Continuity in } u \Longrightarrow [u] = 0 \Longrightarrow s = 0. \end{cases}$$

- All positions of discontinuities are stationary.

# Itay's approach (Picard's iteration)

$$\begin{cases} \partial_t v^n - \partial_x u^n = 0, \\ \partial_t u^n - \partial_x \frac{\kappa}{v^{n-1}} \partial_x u^n = -\partial_x p(v^{n-1}), \text{ (linear heat equation in } u^n) \\ u^n(x,0) = u_0(x), \ v^n(x,0) = v_0(x), \\ v_0 \in C^\sigma(\mathbb{R}), \ u_0 \in C^{1+\sigma}(\mathbb{R}), \ \sigma \in (0,1). \end{cases}$$

Levi's parametrix method (regularity condition requried on $v^{n-1}(x,t)$. This is ensured by the inital condition.)

$\Longrightarrow \exists$ $\mathrm{H}(x,t;y,\tau;v^{n-1})$: The Fundamental solution of

$$\begin{cases} \left(\partial_t - \partial_x \frac{\kappa}{v^{n-1}(x,t)} \partial_x\right) \mathrm{H}(x,t;y,\tau;v^{n-1}) = 0, t \geq \tau, \\ \mathrm{H}(x,\tau;y,\tau;v^{n-1}) = \delta(x-y). \end{cases}$$

$\Longrightarrow$ Convergence of the iterations:

$$\begin{cases} u^n(x,t) = \int_{\mathbb{R}} H(x,t;y,0;v^{n-1}) u_0(y) dy + \int_0^t \int_{\mathbb{R}} \partial_y H(x,t;y,\tau;v^{n-1}) p(v^{n-1}) dy d\tau, \\ v^n(x,t) = v_0(x) - \int_0^t \partial_x u^n(x,\tau) d\tau. \end{cases}$$

- Remark: This becomes a standard procedure to obtain the local existence theory in the $L^2$ energy estimates for to yield the global stability for solutions of the compressible Navier-Stokes developed by Matsumura-Nishda.

# *Functional space for well-posedness*

Similar to Glimm's works on the hyperbolic conservation laws, one may consider the space of the bounded variation (BV) functions as a good candidate for the proper space to accommodate the weak solutions.

• Assumption. The initial data in the $BV \cap L^1$ class.

• **Missing component in the Picard's iteration: Precise pointwise and regularity structure of a fundamental solution** $\mathrm{H}(x, t; y, \tau; v^{n-1})$ **when** $v^{n-1}$ **is a BV function.**

To construct the fundamental solution $\mathrm{H}(x, t; y, \tau; \mu)$ of $\partial_t u - \partial_x \mu(x, t) \partial_x u = 0$ with precise pointwise classical description up to 2nd derivatives with $\mu(x, t)$ in the BV class.

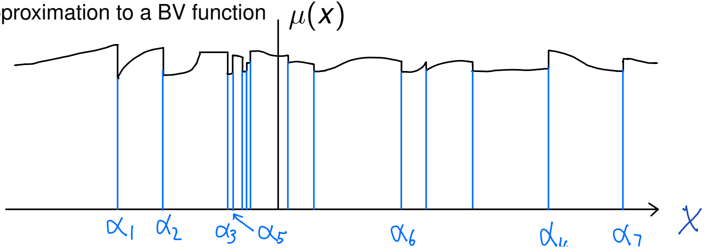
• The Levi's parametrix method can not applied due to lacking regularity property in $\mu(x, t)$.

• De Giorgi, Nash, Moser, etc. had the fundamental solution obtained by **Riesz' representation** and estimated in the Hörlder continuities. It is **NOT SUFFCIENT** for the nonlinear problem.

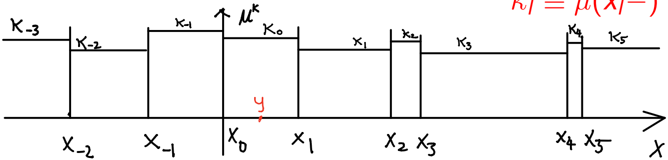
# A functional analysis for the heat equation

$$\partial_t u - \partial_x \mu \partial_x u = 0$$

- Approximation to a BV function




$$\kappa_l \equiv \mu(x_l-)$$




$$\inf_{j\in\mathbb{Z}} |x_j - x_{j-1}| > 0.$$

$$\lim_{k\to\infty} \|\mu - \mu^k\|_\infty = 0.$$

# *Combinatoric solutions*

A procedure to obtain combinatoric the fundamental solution $\mathrm{H}(x,t;y,\mu^k)$ in the weak sense:

$$\begin{cases} \mathrm{H}(x,t;y;\mu^k) \equiv u(x,t), \\ \partial_t u - \partial_x(\mu^k(x)\partial_x u) = 0, \text{ (weak sense)}, \\ u(x,0) = \delta(x-y), \\ \mu^k(x)\text{: A step functions independent of } t. \end{cases}$$

- weak sense $\Longrightarrow$

$$\begin{cases} u(x,t): \text{ Continuous}, \\ \mu^k(x)u_x(x,t): \text{ Continuous}. \end{cases}$$

# *Laplace wave train*

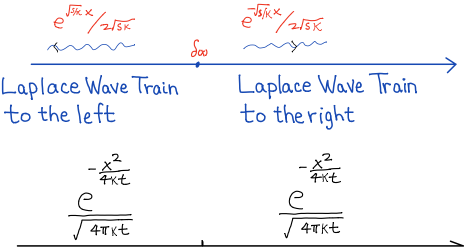
$$\begin{cases} \partial_t u = \kappa \partial_x^2 u, \\ u(x,0) = \delta(x). \end{cases} \iff u(x,t) = \frac{e^{-\frac{x^2}{4\kappa t}}}{\sqrt{4\kappa\pi t}}$$

$\mathbb{L}[u](x,s) \equiv \int_0^\infty e^{-st} u(x,t)dt$. **(Laplace Transform in $t$)**

$$s\mathbb{L}[u](x,s) - \kappa\partial_x^2 \mathbb{L}[u](x,s) = \delta(x) \iff$$

$$\mathbb{L}[u](x,s) = \begin{cases} \frac{e^{-\sqrt{\frac{s}{\kappa}}x}}{2\sqrt{s\kappa}}, & x > 0, \\ \frac{e^{\sqrt{\frac{s}{\kappa}}x}}{2\sqrt{s\kappa}}, & x < 0. \end{cases}$$

**Laplace wave train.**




$$\mathbb{L}^{-1}\left[\frac{e^{-\sqrt{\frac{s}{\kappa}}x}}{2\sqrt{\kappa s}}\right] = \frac{e^{-\frac{x^2}{4\kappa t}}}{\sqrt{4\kappa t}}$$

# *Composite wave train: I*

$$\begin{cases} \partial_t u = \partial_x \mu(x) \partial_x u, \\ u(x,0) = \delta(x-y), \end{cases} \qquad \mu(x) = \begin{cases} \kappa_-, x < 0, \\ \kappa_+, x > 0, \end{cases}$$

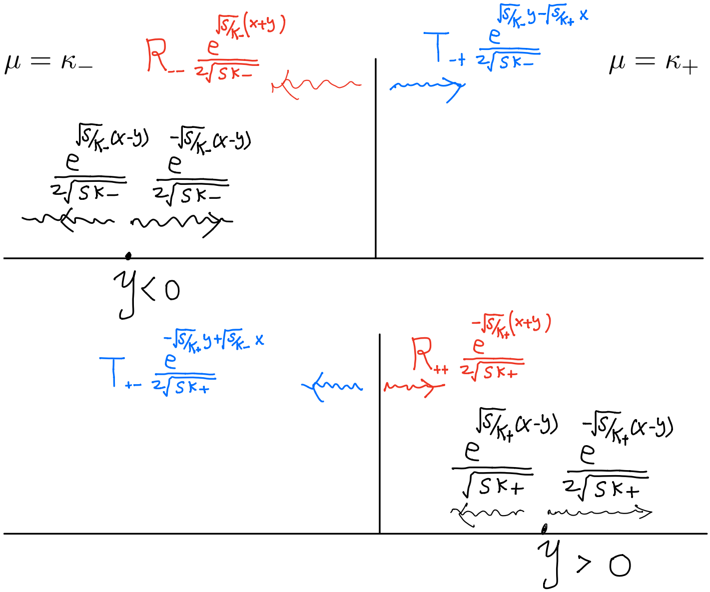
$$\mathbb{L}[u](x,s) = \begin{cases} \frac{e^{-\sqrt{s/\kappa_-}|x-y|}}{2\sqrt{s\kappa_-}} + R_{--}\frac{e^{\sqrt{\frac{s}{kappa_-}}(x+y)}}{2\sqrt{s\kappa_-}}, x, y < 0 \\ T_{-+}\frac{e^{\sqrt{s/\kappa_-}y - \sqrt{s/\kappa_+}x}}{2\sqrt{s\kappa_-}}, x > 0, y < 0, \\ \frac{e^{-\sqrt{s/\kappa_+}|x-y|}}{2\sqrt{s\kappa_+}} + R_{++}\frac{e^{\sqrt{s/\kappa_+}(x+y)}}{2\sqrt{s\kappa_+}}, x, y > 0 \\ T_{+-}\frac{e^{-\sqrt{s/\kappa_+}y + \sqrt{s/\kappa_-}x}}{2\sqrt{s\kappa_+}}, x < 0, y > 0. \end{cases}$$

Weak solution $\Longrightarrow \mathbb{L}[u](x,s)$ and $\mu(x)\partial_x \mathbb{L}[u](x,s)$: Continuous functions.

$$\begin{cases} T_{-+} = 1 + \frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}, \\ R_{--} = \frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}, \end{cases} \qquad \begin{cases} T_{+-} = 1 - \frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}, \\ R_{++} = -\frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}. \end{cases}$$

$T_{\pm\mp} = 1 + O(1)(\kappa_- - \kappa_+)$, $R_{\pm\pm} = O(1)(\kappa_- - \kappa_+)$ when $|\kappa_- - \kappa_+| \ll 1$.

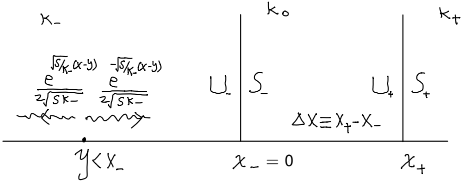
# *Diagram for composite wave train I.*



# Composite wave train II

$$\begin{cases} \partial_t u = \partial_x \mu(x) \partial_x u, \\ u(x,0) = \delta(x-y), \end{cases} \qquad \mu(x) = \begin{cases} \kappa_-, x < x_-, \\ \kappa_0, x \in (x_-, x_+), \\ \kappa_+, x > x_+, \end{cases}$$

- Two diagrams for $y < x_- = 0$: I. **(Resolvent Picture)**

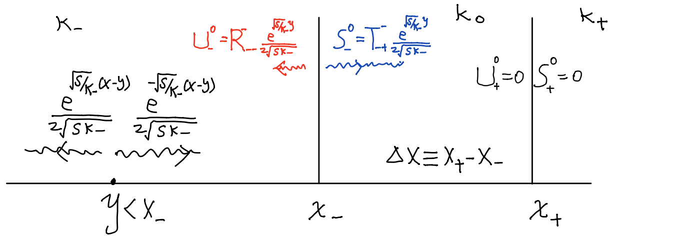


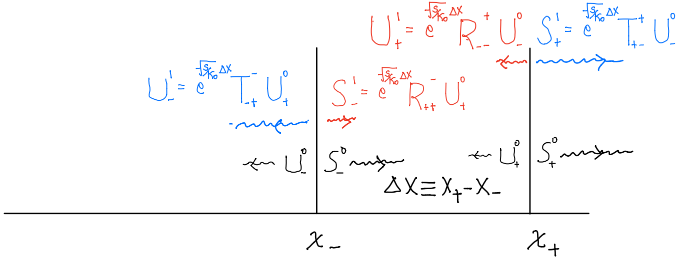

$$\mathbb{L}[u](x,s) = e^{-\sqrt{\frac{s}{\kappa_+}}(x-x_+)} S_+$$

$$\begin{cases} U_- = R^-_{--} \dfrac{e^{\sqrt{s/\kappa_-}y}}{2\sqrt{s\kappa_-}} + T^-_{+-} e^{-\sqrt{s/\kappa_0}\Delta x} U_+, \\ S_- = T^-_{-+} \dfrac{e^{\sqrt{s/\kappa_-}y}}{2\sqrt{s\kappa_-}} + R^-_{++} e^{-\sqrt{s/\kappa_0}\Delta x} U_+, \end{cases} \qquad \begin{cases} U_+ = R^+_{--} e^{-\sqrt{s/\kappa_-0}\Delta x} S_-, \\ S_+ = T^+_{-+} e^{-\sqrt{s/\kappa_-0}\Delta x} S_-. \end{cases}$$

$$\begin{pmatrix} U_- \\ S_- \\ U_+ \\ S_+ \end{pmatrix} = \begin{pmatrix} 1 & 0 & -T^-_{+-} e^{-\sqrt{s/\kappa_0}\Delta x} & 0 \\ 0 & 1 & -R^-_{++} e^{-\sqrt{s/\kappa_0}\Delta x} & 0 \\ 0 & -R^+_{--} e^{-\sqrt{s/\kappa_0}\Delta x} & 1 & 0 \\ 0 & -T^+_{-+} e^{-\sqrt{s/\kappa_0}\Delta x} & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} R^-_{--} \\ T^-_{-+} \\ 0 \\ 0 \end{pmatrix} \frac{e^{\sqrt{\frac{s}{\kappa_-}}y}}{2\sqrt{s\kappa_-}}$$

- Two diagrams for $y < x_-$: II. **(Laplace Wave Train Picture)**



$U_+^{n+1} = e^{-\sqrt{s/\kappa_0}\,\Delta x} R_{--}^{+} U_-^{n}$ $\quad S_+^{n+1} = e^{-\sqrt{s/\kappa_0}\,\Delta x} T_{+-}^{+} U_-^{n}$

$U_-^{n+1} = e^{-\sqrt{s/\kappa_0}\,\Delta x} T_{-+}^{-} U_+^{n}$ $\quad S_-^{n+1} = e^{-\sqrt{s/\kappa_0}\,\Delta x} R_{++}^{-} U_+^{n}$

$U_-^n$ $\quad S_-^n$ $\quad U_+^n$ $\quad S_+^n$

$\Delta x \equiv x_+ - x_-$

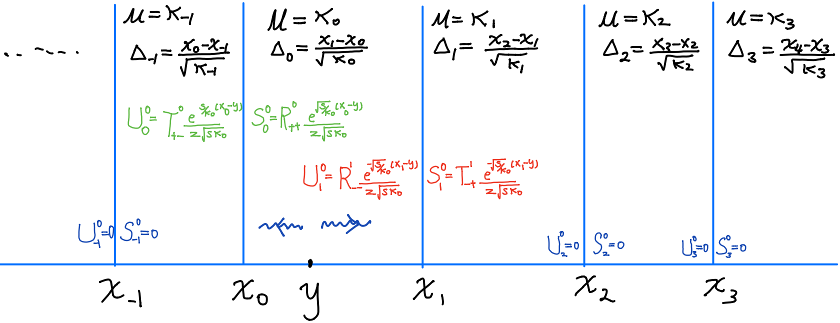
$x_-$ $\quad x_+$

$$\begin{cases} U_- = \sum_{n=0}^{\infty} U_-^n = \sum_{n=0}^{\infty} (T_{-+}^{-} R_{--}^{+} T_{+-}^{-})^n \dfrac{e^{\sqrt{s}\left(\frac{y}{\sqrt{\kappa_-}} - 2n\frac{\Delta x}{\sqrt{\kappa_0}}\right)}}{2\sqrt{s\kappa_-}}, \\ \mathbb{L}[u](x,s) = \dfrac{e^{-\sqrt{s/\kappa_-}|x-y|}}{2\sqrt{s\kappa_-}} \\ + \sum_{n=0}^{\infty} (T_{-+}^{-} R_{--}^{+} T_{+-}^{-})^n \dfrac{e^{\sqrt{s}\left(\frac{y}{\sqrt{\kappa_-}} - 2n\frac{\Delta x}{\sqrt{\kappa_0}} + \frac{(|x-x_-|)}{\sqrt{\kappa_+}}\right)}}{2\sqrt{s\kappa_-}} \quad \text{for } x, y < x_- \end{cases}$$

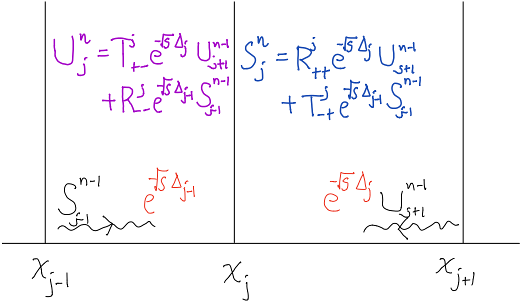
$$u(x,t) = \frac{e^{-\frac{(x-y)^2}{4\kappa_- t}}}{\sqrt{4\pi\kappa_-}} + \sum_{n=0}^{\infty} (T_{-+}^{-} R_{--}^{+} T_{+-}^{-})^n \frac{e^{-\frac{\left(\frac{|x-x_-|+|y-x_-|}{\sqrt{\kappa_-}} + 2n\frac{\Delta x}{\sqrt{\kappa_0}}\right)^2}{4t}}}{\sqrt{4\pi\kappa_-}} \quad \text{for } x, y < x_-$$

# Composite wave train III

$$\begin{cases} \partial_t u = \partial_x \mu(x) \partial_x u, \quad \mu(x) = \kappa_j \text{ for } x \in (x_j, x_{j+1}), \\ u(x,0) = \delta(x-y), \end{cases}$$

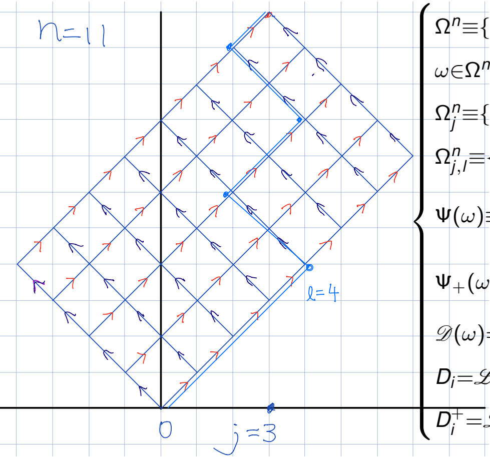


# *The diagram and the difference scheme*




$$\begin{pmatrix} U_j^n \\ S_j^n \end{pmatrix} = \underbrace{\begin{pmatrix} T^j_{+-} & 0 \\ R^j_{++} & 0 \end{pmatrix} e^{-\sqrt{s}\Delta_j}}_{\mathscr{L}_{j+1}\text{: move to left}} \begin{pmatrix} U^{n-1}_{j+1} \\ S^{n-1}_{j+1} \end{pmatrix} + \underbrace{\begin{pmatrix} 0 & R^j_{--} \\ 0 & T^j_{-+} \end{pmatrix} e^{-\sqrt{s}\Delta_{j-1}}}_{\mathscr{R}_{j-1}\text{: move to right}} \begin{pmatrix} U^{n-1}_{j-1} \\ S^{n-1}_{j-1} \end{pmatrix}$$

# *Sample paths, composite Laplace wave train III, and path integral*




$$\Omega^n \equiv \{-1,1\}^n \equiv \{\{a_i\}_{i=1}^n | a_i \in \{-1,1\}\}$$

$\omega \in \Omega^n$: $X(\omega)_k \equiv \sum_{i=1}^k \omega_i$: A path from 0

$$\Omega_j^n \equiv \{\omega \in \Omega^n | \sum_{i=1}^n \omega_i = j\},$$

$\Omega_{j,l}^n \equiv \{\omega \in \Omega_j^n |$ The path $X(\omega)$ changes direction $l$ times$\}$,

$\Psi(\omega) \equiv -\sum_{i=0}^{n-1} \frac{\left| x_{X(\omega)_{i+1}} - x_{X(\omega)_i} \right|}{\sqrt{\kappa_{X(\omega)_i}}}$: phase length

$$\Psi_+(\omega) \equiv -\sum_{i=0}^{n-1} \frac{\left| x_{X(\omega)_{i+1}+1} - x_{X(\omega)_i+1} \right|}{\sqrt{\kappa_{X(\omega)_i+1}}},$$

$$\mathscr{D}(\omega) = D_1 D_2 \cdots D_n,\ \mathscr{D}^+(\omega) = D_1^+ D_2^+ \cdots D_n^+,$$

$D_i = \mathscr{L}_{X(\omega)_i}$ if $\omega(i) = -1$, otherwise $D_i = \mathscr{R}_{X(\omega)_i}$,

$D_i^+ = \mathscr{L}_{X(\omega)_i+1}$ if $\omega(i) = -1$, otherwise $D_i^+ = \mathscr{R}_{X(\omega)_i+1}$.

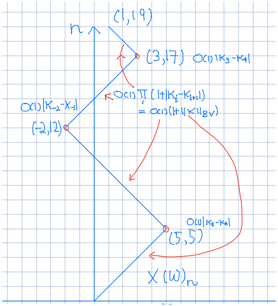
# The scattering data

$$\sum_{n=0}^{\infty}\begin{pmatrix} U_j^n \\ S_j^n \end{pmatrix}$$

$$=\sum_{n=0}^{\infty}\sum_{\omega\in\Omega_j^n}\frac{\mathscr{D}(\omega)e^{\sqrt{s}(\Psi(\omega)-\frac{y-x_0}{\sqrt{\kappa_0}})}}{2\sqrt{s\kappa_0}}+\sum_{n=0}^{\infty}\sum_{\omega\in\Omega_{j-1}^n}\frac{\mathscr{D}^+(\omega)e^{\sqrt{s}\Psi_+(\omega)+\frac{(y-x_1)}{\sqrt{\kappa_0}}}}{2\sqrt{s\kappa_0}}$$

$$=\sum_{l=0}^{\infty}\left(\sum_{n=0}^{\infty}\sum_{\omega\in\Omega_{j,l}^n}\frac{\mathscr{D}(\omega)e^{\sqrt{s}(\Psi(\omega)-\frac{y-x_0}{\sqrt{\kappa_0}})}}{2\sqrt{s\kappa_0}}+\sum_{n=0}^{\infty}\sum_{\omega\in\Omega_{j-1;l}^n}\frac{\mathscr{D}^+(\omega)e^{\sqrt{s}(\Psi_+(\omega)+\frac{(y-x_1)}{\sqrt{\kappa_0}})}}{2\sqrt{s\kappa_0}}\right).$$

- Strength Estimates




$$|\mathscr{D}|\leq O(1)(1+\|\mu\|_{BV})^{l+2}\Pi_{m=1}^{l}|\kappa_{q_m}-\kappa_{q_m+1}|$$

$q_m$ is the turning index.

$$\sum_{n=0}^{\infty}\sum_{\omega\in\Omega_{j,l}^n}|\mathscr{D}|$$

$$\leq O(1)(1+\|\mu\|_{BV})^{l+2}\sum_{q_m\in\mathbb{Z}}\Pi_{m=1}^{l}|\kappa_{q_m}-\kappa_{q_m+1}|$$

$$\leq O(1)(1+\|\mu\|_{BV})^{l+2}\sum_{q_m\in\mathbb{Z}}|\kappa_{q_m}-\kappa_{q_m+1}|$$

$$\leq O(1)(1+\|\mu\|_{BV})^{l+2}\|\mu\|_{BV}^{l}$$

# *Phase length estimate and space-time structure*

- The phase length $\Psi(\omega) < -\frac{1}{2}\int_{x_1}^{x_j}\frac{dx}{\sqrt{\mu(x)}}$.
- The fundamental solution solution. $x \in (x_{j-1}, x_j)$

$$\mathbb{L}[u](x,s) = \sum_{n=0}^{\infty}\left(S_{j-1}^{n} e^{\sqrt{\frac{s}{\kappa_{j-1}}}(x_{j-1}-x)} + U_j^n e^{\sqrt{\frac{s}{\kappa_{j-1}}}(x-x_j)}\right)$$

$$= \sum_{\text{all possible finite paths } X(\omega) \text{ from } y \text{ to } x} C(\omega)\frac{e^{\sqrt{s}\mathscr{I}(\omega)}}{\sqrt{s}}$$

$$\begin{cases} \displaystyle\sum_{\text{all possible finite paths } X(\omega) \text{ from } y \text{ to } x} |C(\omega)| < O(1+\|\mu\|_{BV}), \\ \mathbb{L}^{-1}\left[\frac{e^{\sqrt{s}\mathscr{I}(\omega)}}{2\sqrt{s}}\right] = \frac{e^{-\frac{|\mathscr{I}(\omega)|^2}{4t}}}{\sqrt{4\pi t}} \le O(1)\frac{e^{-\frac{\left(\int_x^y \frac{d\tau}{\sqrt{\mu(\tau)}}\right)^2}{8t}}}{\sqrt{4\pi t}} \end{cases} \Longrightarrow$$

**A CLASSICAL DESCRIPTION OF WEAK SOLUTION**

$$|\partial_x u| \le O(1)\frac{e^{-\frac{\left(\int_x^y \frac{d\tau}{\sqrt{\mu(\tau)}}\right)^2}{8t}}}{t}, \quad |\partial_{xy} u| \le O(1)\frac{e^{-\frac{\left(\int_x^y \frac{d\tau}{\sqrt{\mu(\tau)}}\right)^2}{8t}}}{t^{3/2}}, \cdots$$

# *Weak Limits*

$$\begin{cases} \mathrm{H}(x,t;y;\mu^k) \equiv u(x,t), \\ \partial_t u - \partial_x \mu^k(x) \partial_x u = 0, \\ u(x,0) = \delta(x-y), \\ \ \mu^k(x)\text{: A step functions.} \end{cases}$$

$\lim_{k\to\infty} \|\mu^k - \mu\|_\infty = 0 \Longrightarrow$
$\lim_{k\to\infty} \mathrm{H}(x,t;y,\mu^k)$ : a weak solution of$(\partial_t - \partial_x \mu \partial_x)u = 0$

$$|\partial_x \mathrm{H}(x,t;y,\mu)|, |\partial_y \mathrm{H}(x,t;y,\mu)| \le C_* \frac{e^{-\frac{(x-y)^2}{C_* t}}}{t},$$

$$|\partial_x \partial_y \mathrm{H}(x,t;y,\mu)| \le C_* \frac{e^{-\frac{(x-y)^2}{C_* t}}}{t^{3/2}}, \ \ C_* > 4 \inf \mu.$$

• The Green's function $\mathrm{H}(x,t;y,\mu)$ can be improved to $\mathrm{H}(x,t;y,\tau;\mu)$ with $\mu = \mu(x,t)$ so that the standard iteration works.

$\mathrm{H}(x,t;y,\tau;\mu)$: The Green's function for space-time dependent problem with a BV-function $\mu$.

# The local existence of weak solution

- The standard iteration

$$\begin{cases} \partial_t v^n - \partial_x u^n = 0, \\ \partial_t u^n + \partial_x p(v^{n-1}) - \partial_x \frac{\kappa}{v^{n-1}} \partial_x u^n = 0, \\ u^n(x,0) = u_0(x), \ v^n(x,0) = 1 + v_0(x). \end{cases}$$

($u_0$ and $v_0$ are $BV$ step functions).

$\mathfrak{D} \equiv \{x \in \mathbb{R} | (v_0, u_0)|_{x-}^{x+} \neq (0,0)\}$.

- Four identities.

$$\begin{aligned}
(U^{n+1} - U^n)_x(x,t) &= \int_{\mathbb{R}} \Big(\mathrm{H}_x(x,t;y,0;\kappa/V^n) - \mathrm{H}_x(x,t;y,0;\kappa/V^{n-1})\Big) u_0(y) dy \\
&\quad - \int_{\mathbb{R}} \int_0^t \Big( (\mathrm{H}_x(x,t;y,\tau;\kappa/V^n) - \mathrm{H}_x(x,t;y,\tau;\kappa/V^{n-1})) p(V^n)_y(y,t) \\
&\qquad - \mathrm{H}_x(x,t;y,\tau;\kappa/V^{n-1})(p(V^n) - p(V^{n-1}))_y(y,t)\Big) dy d\tau \\
&\quad - \sum_{z \in \mathfrak{D}} \int_0^t \Big(\mathrm{H}_x(x,t;z,\tau;\kappa/V^n) - \mathrm{H}_x(x,t;z,\tau;\kappa/V^{n-1})\Big) p(V^n)(\cdot,t)\Big|_{z-}^{z+} d\tau \\
&\quad - \sum_{z \in \mathfrak{D}} \int_0^t \mathrm{H}_x(x,t;z,\tau;\kappa/V^{n-1}) \Big(p(V^n) - p(V^{n-1})\Big)(\cdot,t)\Big|_{z-}^{z+} d\tau \\
&\quad - \int_0^t \int_{\mathbb{R}\setminus\mathfrak{D}} \int_\tau^t \Big(\mathrm{H}_{xy}(x,t;y,\tau;\kappa/V^n) - \mathrm{H}_{xy}(x,t;y,\tau;\kappa/V^{n-1})\Big)(p'(V^n)U_y^n)(y,\sigma) d\sigma dy d\tau \\
&\quad - \int_0^t \int_{\mathbb{R}\setminus\mathfrak{D}} \int_\tau^t \mathrm{H}_{xy}(x,t;y,\tau;\kappa/V^{n-1}) \Big(p'(V^n)U_y^n - p'(V^{n-1})U_y^{n-1}\Big)(y,\sigma) d\sigma dy d\tau.
\end{aligned}$$

$$\|(V^{n+1} - V^n)(\cdot, t)\|_{L^1} = \int_{\mathbb{R}} \Big| \int_0^t (U^{n+1} - U^n)_x(x, \tau_1) d\tau_1 \Big| dx$$

$$= \kappa \int_{\mathbb{R}} \Big| \int_0^t \int_{\mathbb{R}\setminus\mathfrak{D}} \int_\tau^t \mathrm{H}_{xy}(x, \tau_1; y, \tau; \kappa/V^n) \Big( p(V^{n-1}) - p(V^{n-1}) + \frac{(V^n - V^{n-1})U_y^n}{(1 + V^n)(1 + V^{n-1})} \Big) d\tau_1 dy d\tau \Big| dx,$$

$$\int_{\mathbb{R}\setminus\mathfrak{D}} |(V_x^{n+1} - V_x^n)(\cdot, t)| dx =$$

$$\kappa \int_{\mathbb{R}\setminus\mathfrak{D}} \Big| \int_0^t \int_{\mathbb{R}\setminus\mathfrak{D}} \int_\tau^t \mathrm{H}_{xxy}(x, \tau_1; y, \tau; \kappa/V^n) \Big( p(V^n) - p(V^{n-1}) + \frac{(V^n - V^{n-1})U_y^n}{(1 + V^n)(1 + V^{n-1})} \Big) d\tau_1 dy d\tau \Big| dx.$$

$$\partial_t (V^{n+1} - V^n)(\cdot, t)\Big|_{x=z-}^{x=z+} = \frac{(V^n - V^{n-1})}{\kappa}(\cdot, t)\Big|_{x=z-}^{x=z+} \Big( \kappa \frac{U_x^{n+1}}{V^n} - p(V^n) \Big)$$

$$+ \frac{V^{n-1}}{\kappa}(\cdot, t)\Big|_{x=z-}^{x=z+} \Big( \kappa \frac{U_x^{n+1}}{V^n} - \kappa \frac{U_x^n}{V^{n-1}} - p(V^n) + p(V^{n-1}) \Big) + \frac{1}{\kappa} \Big( V^n p(V^n) - V^{n-1} p(V^{n-1}) \Big)(\cdot, t)\Big|_{x=z-}^{x=z+}.$$

# The metric space

$$\begin{cases} |||f|||_\infty \equiv \sup_{\sigma\in(0,t_\sharp)} \|f(\cdot,\sigma)\|_\infty, \\ |||f|||_{BV} \equiv \sup_{\sigma\in(0,t_\sharp)} \|f(\cdot,\sigma)\|_{BV}, \\ |||f|||_{L^1} \equiv \sup_{\sigma\in(0,t_\sharp)} \|f(\cdot,\sigma)\|_{L^1}, \end{cases} \quad |||f|||_{\mathfrak{D}} \equiv \sup_{\sigma\in(0,t_\sharp)} \sup_{z\in\mathfrak{D}} \left| \frac{f(\cdot,\sigma)\Big|_{z-}^{z+}}{v_0(\cdot)\Big|_{z-}^{z+}} \right|.$$

- The estimates

$$|||\sqrt{t}U_x^{n+1}|||_\infty \leq O(1)\delta,$$

$$|||\frac{\sqrt{t}}{|\log t|}(U^{n+1}-U^n)_x|||_\infty \leq O(1)\delta\Big(|||V^n - V^{n-1}|||_\infty + \frac{|||V^n - V^{n-1}|||_{BV}}{|\log t_\sharp|} + \sqrt{t_\sharp}|||V^n - V^{n-1}|||_{L^1} + t_\sharp|||\frac{\sqrt{t}}{|\log t|}(U^n - U^{n-1})_x|||_\infty\Big),$$

$$|||V^{n+1} - V^n|||_{L^1} \leq O(1)\sqrt{t_\sharp}(1+O(1)\delta)|||V^n - V^{n-1}|||_{L^1},$$

$$\int_{\mathbb{R}\setminus\mathfrak{D}} |(V^{n+1}-V^n)_x(x,t)|dx \leq O(1)|||V^n - V^{n-1}|||_{L^1}(\sqrt{t} + |||\sqrt{t}U_x^n|||_\infty),$$

$$|||V^{n+1} - V^n|||_{\mathfrak{D}} \leq O(1)\delta\sqrt{t_\sharp}|||V^n - V^{n-1}|||_{\mathfrak{D}} + O(1)|\log t_\sharp|\sqrt{t_\sharp}|||\frac{\sqrt{t}}{|\log t|}(U^n - U^{n-1})_x|||_\infty + O(1)\sqrt{t_\sharp}|||\sqrt{t}U_x^{n-1}|||_\infty|||V^n - V^{n-1}|||_\infty.$$

**Standard Iteration scheme converges for $t \leq t_\sharp \ll 1$.**

# *Conclusion I. Local existence*

$(\boldsymbol{B}_\delta, d)$: Metric space for initial data.

$$\begin{cases} \boldsymbol{B}_\delta \equiv \{(v_0, u_0) : \|v_0 - 1\|_{L^1} + \|v_0\|_{BV} + \|u_0\|_{L^1} + \|u_0\|_{BV} \leq \delta, \\ d((v_0^a, u_0^a), (v_0^b, u_0^b)) \equiv \|v_0^a - v_0^b|_{L^1} + \|u_0^a - u_0^b\|_{BV}. \end{cases}$$

$\boldsymbol{S}^{t_\sharp} \equiv \{(v_0, u_0) \in \boldsymbol{B}_\delta :$ The initial data admits solution for $t \in (0, t_\sharp)\}$

**Theorem.** There exist $\delta$ and $t_\sharp > 0$ such that $\boldsymbol{S}^{t_\sharp}$ is dense in $(\boldsymbol{B}_\delta, d)$.

# *Hadamard's well-posedness problem*

- One needs to require $0 < \gamma < e$ for $p(v) = v^{-\gamma}$.
- Green's function $\mathbb{G}(x,t;\alpha,\beta)$, $\alpha,\beta > 0$, of a linear system:

$$\begin{pmatrix} \partial_t & -\partial_x \\ -\beta^2\partial_x & \partial_t - \alpha\partial_x^2 \end{pmatrix} \mathbb{G}(x,t;\alpha,\beta) = \vec{0}, \ \mathbb{G}(x,0;\alpha,\beta) = \delta(x)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

- Singular-Regular decomposition of $\mathbb{G}(x,t;\alpha,\beta)$:

$$\begin{cases} \mathbb{G} \equiv \mathbb{G}^* + \mathbb{G}^\sharp, \\ \mathbb{G}^* = \mathbb{G}^{*,1} + \mathbb{G}^{*,2}, \end{cases}$$

$$\begin{cases} \mathbb{G}^* = \mathbb{G}^{*,1} + \mathbb{G}^{*,2}, \\ \left|\partial_t^i\left(\left(\left(\mathbb{G}^{*,1} - \delta(x)e^{-\frac{\beta^2}{\alpha}t}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right)\right)_x + \frac{e^{-\frac{\beta^2}{\alpha}t}}{\alpha}\delta(x)\begin{pmatrix} 0 & 1 \\ \beta^2 & 0 \end{pmatrix}\right)\right| \le O(1)e^{-\sigma_*|x|}, \\ \mathbb{G}^{*,2}(x,t;\alpha,\beta) = O(1)e^{-\sigma^* t - \sigma_0|x|} + \begin{cases} \frac{1}{\sqrt{4\pi\alpha t}}e^{-\frac{x^2}{4\alpha t}}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, & 0 < t < 1; \\ 0, \ t \ge 1, \end{cases} \\ |\partial_x^k \mathbb{G}^\sharp| \le O(1)\frac{e^{-\frac{x^2}{C_*(t+1)}}}{\sqrt{(t+1)^{k+1}}} \text{ for } k = 0,1,2. \end{cases}$$

# An integral representation of the weak solution

Fixed $(x,t)$ and $\bar{v} \equiv v(x,t)$. Take $\bar{\alpha} = \kappa/\bar{v}$ and $\bar{\beta} = \sqrt{-p'(\bar{v})}$.
Consider $\int_0^t \int_{\mathbb{R}} \mathbb{G}(x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \begin{pmatrix} \partial_\tau v - \partial_y u \\ \partial_\tau u + \partial_y p(v) - \kappa \partial_y \frac{\partial_y u}{v} \end{pmatrix} dy d\tau = \vec{0}.$

$$
\begin{aligned}
\bar{v} = & \int_{\mathbb{R}} \mathbb{G}_{11}(x-y, \bar{\beta}, \bar{\alpha}) v_0(y) + \mathbb{G}_{12}(x-y, \bar{\beta}, \bar{\alpha}) u_0(y) dy \\
& + \int_0^t \int_{\mathbb{R}} \partial_x \mathbb{G}_{12}(x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \Big( -(p(v) - p(\bar{v}) - p'(\bar{v})(v - \bar{v})) \Big)(y,\tau) dy d\tau \\
& \qquad + \int_0^t \int_{\mathbb{R}} \partial_x \mathbb{G}_{12}(x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \Big( \log v - \frac{v}{\bar{v}} \Big)_\tau dy d\tau.
\end{aligned}
$$

$$\bar v = \Big( e^{-\frac{\bar\beta^2}{\bar\alpha}t}\big(\bar v\log\frac{v_0(x)}{\bar v} + \bar v - v_0(x)\big) + \int_{\mathbb R} \mathbb G_{11}(x-y,t;\bar\alpha,\bar\beta)v_0(y)dy\Big)$$
$$+ \int_{\mathbb R} \mathbb G_{12}(x-y,t;\bar\alpha,\bar\beta)u_0(y)dy$$
$$+\kappa\int_{\mathbb R}\Big(\mathrm D_{12}(x-y,0;\bar\alpha,\bar\beta)\Big(\log\big(\frac{v(y,t)}{\bar v}\big)-\frac{v(y,t)-\bar v}{\bar v}\Big)-\mathrm D_{12}(x-y,t;\bar\alpha,\bar\beta)\Big(\log\big(\frac{v_0(y)}{\bar v}\big)-\frac{v_0(y)-\bar v}{\bar v}\Big)\Big)\,dy$$
$$+\int_0^t \frac{1}{\bar\alpha}e^{-\frac{\bar\beta^2}{\bar\alpha}(t-\tau)}\big(p(v(x,\tau))-p(\bar v)-p'(\bar v)(v(x,\tau)-\bar v)\big)d\tau$$
$$+\int_0^t\int_{\mathbb R}\partial_y\Big(\mathbb G_{12}-\mathbb G_{12}^{*,1}\Big)(x-y,t-\tau;\bar\alpha,\bar\beta)\big(p(v(y,\tau))-p(\bar v)-p'(\bar v)(v(y,\tau)-\bar v)\big)dyd\tau$$
$$-\int_0^t\int_{\mathbb R}\partial_y\Big(\mathbb G_{12}-\mathbb G_{12}^{*,1}\Big)(x-y,t-\tau;\bar\alpha,\bar\beta)\kappa\big(\frac{u_y(y,\tau)}{v(y,\tau)}-\frac{u_y(y,\tau)}{\bar v}\big)dyd\tau$$
$$+\int_0^t\frac{\bar\beta^2}{\bar\alpha}e^{-\frac{\bar\beta^2}{\bar\alpha}(t-\tau)}\big(\bar v\log\big(\frac{v(x,\tau)}{\bar v}\big)-(v(x,\tau)-\bar v)\big)d\tau$$
$$+\int_0^t\int_{\mathbb R}O(1)e^{-\sigma_*(t-\tau)-\sigma_0|x-y|}\Big(\log\big(\frac{v(y,\tau)}{\bar v}\big)-\frac{v(y,\tau)-\bar v}{\bar v}\Big)dyd\tau.$$

$$u_x = \int_{\mathbb R}\mathrm H_x(x,t;y,0;\kappa/v)u_0(y)dy - \int_0^t\int_{\mathbb R}\mathrm H_x(x,t;y,\tau;\kappa/v)p(v(y,t))_y\,dyd\tau$$
$$-\sum_{z\in\mathfrak D}\int_0^t \mathrm H_x(x,t;z,\tau;\kappa/v)p(v(,t))d\tau\Big|_{z-}^{z+}+\int_0^t\int_{\mathbb R\setminus\mathfrak D}\int_\tau^t \mathrm H_{xy}(x,t;y,\tau;\kappa/v)(p'(v(y,\sigma))u_y(y,\sigma)d\sigma dyd\tau.$$

$$\sup_{t>0} \frac{d}{d\bar{v}} e^{-\frac{\bar{\beta}^2}{\bar{\alpha}} t} \left(\bar{v} \log \frac{v_0(x)}{\bar{v}} + \bar{v}\right) = \sup_{t>0} \frac{d}{d\bar{v}} e^{\frac{p'(\bar{v})\bar{v}}{\kappa} t} \left(\bar{v} \log \frac{v_0(x)}{\bar{v}} + \bar{v}\right) < \gamma/e < 1 \text{ if } \gamma < e.$$

$\Longrightarrow$

**Theorem.** For any two initial data $(v_0^a, u_0^a)$ and $(v_0^b, u_0^b) \in \boldsymbol{S}^{t_\sharp}$, the solutions $(v^a, u^a)$ and $(v^b, u^b)$ satisfy

$$\|(v^a - v^b)(\cdot, t)\|_{L^1} + \|(u^a - u^b)(\cdot, t)\|_{L^1} \le O(1)\Big(\|v_0^a - v_0^b\|_{L^1} + \|u_0^a - u_0^b\|_{L^1}\Big).$$

**Hadamard's Well-Posedness** problem for NS in $\|\cdot\|_{BV} + \|\cdot\|_{L^1}$ is **SOLVED**.